

AGENTIC PAYMENT RECOVERY

Repechage

**Agentic Payment Recovery for
Failed Razorpay Checkouts**

AI reasons. Policy decides. Execution is controlled.

**Every failed checkout is a customer
who tried to pay you.**

Repechage brings them back.

**Every failed checkout is a customer
who tried to pay you.**

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From failed payment to controlled recovery

The revenue loss is already in the funnel.

Customers are ready to pay. The checkout fails
—and intent evaporates.

₹98,24,112

transaction value

66.6%

flagged at-risk • ₹65,38,889

438

genuinely recoverable events

`card_declined` • `insufficient_funds` • `otp_timeout` • `network_error`

1,000 failed-payment events

Same trigger. Same action. Wrong outcome.

NAIVE AUTOMATION

Wastes attempts
One response for every failure

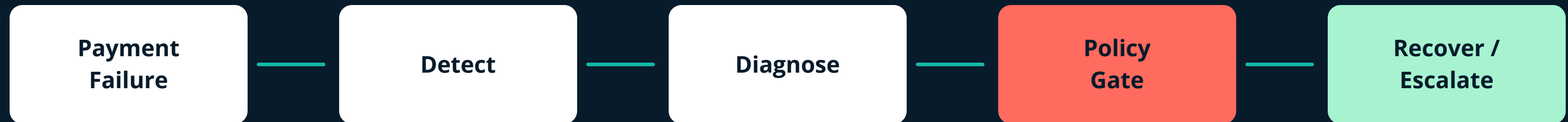
Annoys customers
Irrelevant, repeated actions

Misses recoveries
No diagnosis of the cause

Never knows when to stop
No explicit stopping criteria

Reason broadly. Act narrowly.

AI reasons. Policy decides. Execution is controlled.



A bounded decision, not an open-ended agent.

Every decision → audit trail

01 DETECT

Deterministic rules on status + failure_reason

02 DIAGNOSE

LLM Recovery Agent reads rich transaction context

03 POLICY GATE

Safety and business rules hold final authority

recover_now

send_payment_link

wait_and_retry

escalate_to_merchant

stop

The policy gate is the authority.

WHEN IN DOUBT → ESCALATE, DO NOT EXECUTE

Attempts ≥ 3

STOP

Already recovered

STOP

Amount $> ₹5,000$

ESCALATE

Max 10 real actions / run

HARD CAP

Live execution

DEFAULT OFF

Invalid / low-confidence

GATED

Two models. One hard boundary.

DETERMINISTIC DETECTOR

status • failure_reason • attempts
Explainable risk classification
Zero LLM cost

LLM RECOVERY AGENT

Rich transaction context
Diagnosis + action + confidence
Structured recommendation

POLICY GATE
and truth. Neither c

Structured in. Auditable out.

```
{  
  "diagnosis": "card_declined",  
  "recovery_probability": 0.82,  
  "recommended_action": "send_payment_link",  
  "reason": "New link can recover payment",  
  "confidence": 0.91  
}
```

SCHEMA-VALIDATED → POST-FILTERED → POLICY-GATED

AUDIT SURFACE

transactions
detection_results
agent_decisions
recovery_attempts
logs • webhooks • merchants

From merchant signal to accountable action.

PostgreSQL on Neon • 8 tables • Alembic migrations

FRONTEND

Dashboard • Analytics • Audit • Developers • Security

API LAYER

FastAPI • pages • auth • webhooks • dashboard

DECISION

Detect → LLM Agent → Policy Gate

DATABASE

Transactions + execution lifecycle

One failed checkout. A controlled path back.

Recovery executes only when policy approves.



Capture → reason → authorize → execute → explain

Safer interventions. Lower total recovery.

73.0% AI HIT-RATE
vs 66.2% baseline



ENGINEERING LESSON / Precision improved, but fewer actions meant less net revenue. Optimize for the right opportunities—not simply fewer interventions.

THE NEXT RECOVERY SHOULD BE
EXPLAINABLE, SAFE, AND WORTH IT.

Recover revenue. Keep control.

AI reasoning + strict policy control + transparent audit.

LET'S DISCUSS →